

## Document 4 -- Additional Information

Aran Technologies | [aranrobotics@gmail.com](mailto:aranrobotics@gmail.com) | +91 72888 40612

### I. PHASE I TESTING -- COMPLETED

All Phase I systems have been tested and verified. The five-drone swarm simulation on ROS2 Jazzy and Gazebo Harmonic has demonstrated all core MBC-3 requirements: real-time ASP generation, bully-protocol leader election, graceful degradation on drone loss, and sector reassignment -- 344 radar tracks logged with four surviving drones sustaining full mission continuity.

A pre-recorded 4-minute single-drone ISR mission video (1920x1080, v12-MPC-v5) is ready for Phase I presentation: 11-waypoint survey, primary target orbit locked at 50 m radius +/-0.5 m, RTL, and 3D occupancy map save -- verified clean exit. Broadband and drone-to-ground communication links have been tested and confirmed operational.

### II. INDIGENISATION BREAKDOWN

| Sub-system                        | Indigenous Content                                 |
|-----------------------------------|----------------------------------------------------|
| Hexacopter airframe               | Indigenous design and local fabrication            |
| Antenna panels (6 x per drone)    | Indigenous PCB design                              |
| AI software stack (CFAR, RF, LLM) | 100% indigenous (Python / C++, open-source models) |
| GCS dashboard (Flask / SocketIO)  | 100% indigenous                                    |
| ROS2 swarm coordination stack     | 100% indigenous                                    |
| Flight controller firmware        | Custom STM32 PCB -- designed from scratch          |
| Compute module (Jetson)           | COTS -- imported                                   |
| Radar frontend (AWR1843)          | COTS -- imported                                   |
| Mesh radio (Doodle Labs)          | COTS -- imported                                   |

Estimated indigenisation: >= 55% by mission-criticality weighting -- meets MBC-3 S.2.25 requirement.

### III. TEAM

Aran Technologies is a three-member engineering startup building indigenous defence UAS systems from the ground up.

Kishore Udhayakumar | Founder & CEO -- Mechanical Engineering. Leads product development, mechanical design, thermal engineering, and business strategy. Drives go-to-market planning and direct engagement with Indian defence end-users.

Vegi Tejo Bhargav | Co-Founder & CTO -- Mechanical Engineering. Manages project execution, hexacopter airframe fabrication, additive manufacturing, procurement, and hardware integration roadmap.

L. Harsha Vardhan Naidu | Product Head, Electronics -- EEE. Designed the complete autonomous mission stack: PX4 SITL pipeline, ROS2 swarm middleware, GCS dashboard, STM32 flight controller firmware, and three-layer onboard AI pipeline (CFAR -> Random Forest -> LLM).

Point of Contact: [aranrobotics@gmail.com](mailto:aranrobotics@gmail.com) | +91 72888 40612

### IV. PHASE I READINESS

Simulation stack, GCS dashboard, graceful degradation demonstration, and pre-recorded mission video are complete and ready for live presentation at New Delhi, 13-24 July 2026.